SLEEP 02

Glycine

The simplest amino acid (aminoacetic acid)



CLASS	STUDIED FOR	HUMAN EVIDENCE	US STATUS
Amino acid	Sleep, next-day fatigue	Very small human studies	Dietary supplement · not a sleep drug

STUDIED FOR

Next-day fatigue	Sleep
HUMAN	HUMAN + ANIMAL

Sleep research is small. A 2026 review found 3 glycine sleep trials and called for larger trials with standard sleep measures.

HOW IT WORKS IN THE BODY

Glycine → NMDA receptors → Body clock (SCN) → Non-REM sleep

- **In rats,** glycine by mouth increased non-REM sleep and lowered body temperature
- **Traced to NMDA receptors** in the brain's body-clock center, in rats
- **In people,** the mechanism is not established

EVIDENCE & WHAT'S KNOWN

EVIDENCE

CASE REPORTS
LEVEL 2 OF 5

- 3 small sleep trials (2026 scoping review)
- Crossover study: 7 men analyzed
- Sleep itself not recorded in that study

NOT YET ESTABLISHED

- Long-term safety
- Side-effect profile
- Half-life in people
- Effect on recorded sleep

QUICK PRACTICAL INFO

HALF-LIFE	ROUTES STUDIED
Not established in people	By mouth, before bed (studies)
PHARMACOKINETICS	**SOLD AS**
Limited human data	Dietary supplement
MONITORED IN TRIALS	**SPORT (WADA)**
Sleepiness, fatigue, vigilance tests	Not on the 2026 list

REGULATORY

FDA	Not approved for sleep; sold as a supplement
FDA	Approved only as a surgical irrigant
WADA	Not on the 2026 list

REFERENCES

1. Front Neurol, Bannai et al., glycine after sleep restriction, Apr 2012
2. Neuropsychopharmacology, Kawai et al., glycine and sleep in rats, May 2015
3. J Diet Suppl, Fan et al., scoping review of supplements and sleep, Sep 2026

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